Classical Monetary Model

Assumptions

- Representative Agents with rational expectations.
- Perfect Competition in the Market.
- No frictions: Fully flexible prices in all markets.
- Very limited role to money (Money = unit of account).

Households

- The representative household maximize the objective function:

$$E_0 \sum_{t=0}^{\infty} \beta^t U(C_t, N_t)$$

  - C_t is the quantity consumed. N_t are worked hours or employment.

  The period utility is assumed to be continuously and differentiable. And the marginal utility of consumption is assumed to be positive and nondecreasing, while the marginal disutility of labor is positive and nondecreasing.

$$U_{c,t} = \frac{\partial U(C_t, N_t)}{\partial C_t} > 0;\ U_{cc,t} = \frac{\partial^2 U(C_t, N_t)}{\partial C_t^2} \leq 0$$

$$U_{n,t} = \frac{\partial U(C_t, N_t)}{\partial N_t} < 0;\ U_{nn,t} = \frac{\partial^2 U(C_t, N_t)}{\partial C_t^2} \leq 0$$

- Maximization is subject to budgets constraints given by:

$$P_t C_t + Q_t B_t \leq B_{t-1} + W_t N_t - T_t$$

  - P_t is the price of the consumption good. W_t is the nominal wage. B_t is the quantity of one-period discounted bonds purchased in period t and maturating in period $t+1$. T_t represents lump-sum additions or subtractions to period income.
- Also, it's subject to a solvency constraint:

$$\lim_{T \to \infty} E_t\{B_T\} \geq 0$$

  This prevents Ponzi-type schemes.

Optimal consumption and Labor Supply

- Solving:

$$L = E_0 \sum_{t=0}^{\infty} \beta^t [U(C_t, N_t) + \lambda_t (B_{t-1} + W_t N_t - T_t - P_t C_t - Q_t B_t)]$$

  - λ_t represents the marginal utility of wealth (or income) in period t.

$$\frac{\partial L}{\partial C_t} \to \beta^t \left[\frac{\partial U(C_t, N_t)}{\partial C_t} - \lambda_t P_t\right] = 0 \to \frac{\partial U(C_t, N_t)}{\partial C_t} \cdot \frac{1}{P_t} = \lambda_t \quad (1)$$

$$\frac{\partial L}{\partial N_t} \to \beta^t \left[\frac{\partial U(C_t, N_t)}{\partial N_t} + \lambda_t W_t\right] = 0 \to \frac{\partial U(C_t, N_t)}{\partial N_t} \cdot \frac{1}{W_t} = -\lambda_t \quad (2)$$

$$\frac{\partial L}{\partial B_t} \to -\beta^t \lambda_t Q_t + \beta^{t+1} E_t[\lambda_{t+1}] = 0 \to \lambda_t Q_t = \beta E_t[\lambda_{t+1}] \quad (3)$$

- Substitute (1) into (2):

$$\frac{\partial U(C_t, N_t)}{\partial C_t} \cdot \frac{1}{P_t} = -\frac{\partial U(C_t, N_t)}{\partial N_t} \cdot \frac{1}{W_t}$$

$$-\frac{U_n(C_t, N_t)}{U_c(C_t, N_t)} = \frac{W_t}{P_t}$$

Note: This is the optimal intra-temporal condition (Intra-temporal Euler equation). The household equates its marginal rate of substitutions (MRS) between labor and consumption to the real wage. Intuitive, the marginal cost of working an additional hour is exactly compensated by the marginal benefit obtained from consuming the goods that can be purchased with the additional wage.

- Substituting (1) into (3) for both periods t and $t+1$:

$$\frac{\partial U(C_t, N_t)}{\partial C_t} \cdot \frac{1}{P_t} Q_t = \beta E_t \left[\frac{\partial U(C_{t+1}, N_{t+1})}{\partial C_{t+1}} \cdot \frac{1}{P_{t+1}} \right]$$

$$Q_t = \beta E_t \left[\frac{U_c(C_{t+1}, N_{t+1})}{U_C(C_t, N_t)} \cdot \frac{P_t}{P_{t+1}} \right]$$

$$U_C(C_t, N_t) = \beta E_t [U_c(C_{t+1}, N_{t+1})(1 + i_t)(1 + \pi_{t+1})^{-1}]$$

Note: This condition determines the optimal path of consumption and saving for the household. The household must be indifferent at the margin between consuming an additional unit of wealth today or using those funds to purchase a bond. The price of the bond today Q_t must equal the expected discounted marginal value of the payoff tomorrow. $\beta E_t \left[\frac{U_c(C_{t+1}, N_{t+1})}{U_C(C_t, N_t)} \right]$ is the stochastic discount factor and $\frac{P_t}{P_{t+1}}$ accounts for inflation $\left(\frac{1}{\pi_{t+1}} \right)$.

- Assuming that the period utility takes the form:

$$U(C_t, N_t) = \frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\varphi}}{1+\varphi}$$

The consumer's optimality conditions thus become:

$$N_t^{\varphi} C_t^{\sigma} = \frac{W_t}{P_t}$$

$$C_t^{-\sigma} = \beta E_t [C_{t+1}^{-\sigma}(1 + i_t)(1 + \pi_{t+1})^{-1}]$$

Firm

- The representative firm is described by a production function given by:

$$Y_t = A_t N_t^{1-\alpha}$$

  - A_t represents the level of technology that evolves exogenously according to some stochastic process.
- Each period the firm maximizes profits

$$P_t Y_t - W_t N_t$$

  - Assumption: the firm takes prices and wages as given.

Optimal Demand for Labor

- Solving:

$$\max_{N_t} \Pi_t = P_t Y_t - W_t N_t$$

$$s.t. \qquad Y_t = A_t N_t^{1-\alpha}$$

- Profit maximization function:

$$\max_{N_t} \Pi_t = P_t A_t N_t^{1-\alpha} - W_t N_t$$

$$\frac{\partial \Pi_t}{\partial N_t} \rightarrow (1-\alpha) P_t A_t N_t^{-\alpha} - W_t = 0$$

$$\frac{W_t}{P_t} = (1-\alpha) A_t N_t^{-\alpha}$$

Note: This equation represents the optimal labor demand condition. The firm hires labor up to the point where its marginal product equals the real wage. Equivalently, the marginal cost $\frac{W_t}{(1-\alpha)A_t N_t^{-\alpha}}$ must be equal to the price P_t.

Equilibrium

The intra-temporal equations solve for the real variables $\{Y_t, N_t, \frac{W_t}{P_t}\}$ given the exogenous shock.

- The good market clearing condition is given by:

$$Y_t = C_t$$

- The production function:

$$Y_t = A_t N_t^{1-\alpha}$$

$$A_t = \rho_a A_{t-1} + \varepsilon_t^a$$

- Euler Equation:

$$C_t^{-\sigma} = \beta E_t [C_{t+1}^{-\sigma} (1+i_t)(1+\pi_{t+1})^{-1}]$$

- Labor Supply:

$$N_t^{\varphi} C_t^{\sigma} = \frac{W_t}{P_t} \quad \rightarrow \quad \frac{W_t}{P_t} = N_t^{\varphi} Y_t^{\sigma}$$

- Labor Demand:

$$\frac{W_t}{P_t} = (1-\alpha)A_t N_t^{-\alpha}$$

- Equating labor demand and labor supply, we obtain:

$$N_t^{\varphi} Y_t^{\sigma} = (1-\alpha)A_t N_t^{-\alpha}$$

Replacing N_t in the previous equation:

$$\left(\frac{Y_t}{A_t}\right)^{\frac{\varphi}{1-\alpha}} Y_t^{\sigma} = (1-\alpha)A_t \left(\frac{Y_t}{A_t}\right)^{\frac{-\alpha}{1-\alpha}}$$

$$Y_t^{\left(\frac{\varphi+(1-\alpha)\sigma+\alpha}{1-\alpha}\right)} = (1-\alpha)A_t^{\left(\frac{1+\varphi}{1-\alpha}\right)}$$

$$Y_t = \left((1-\alpha)A_t^{\left(\frac{1+\varphi}{1-\alpha}\right)}\right)^{\frac{1-\alpha}{\varphi+(1-\alpha)\sigma+\alpha}}$$

Thus, N_t is given by:

$$N_t = \left(\frac{Y_t}{A_t}\right)^{\frac{1}{1-\alpha}}$$

$$N_t = \left((1-\alpha)A_t^{\left(\frac{1+\varphi}{1-\alpha}\right)}\right)^{\frac{1}{\varphi+(1-\alpha)\sigma+\alpha}} . A_t^{\frac{-1}{1-\alpha}}$$

- Replacing N_t and Y_t in the labor supply:

$$\frac{W_t}{P_t} = \left(\left((1-\alpha)A_t^{\left(\frac{1+\varphi}{1-\alpha}\right)}\right)^{\frac{1}{\varphi+(1-\alpha)\sigma+\alpha}} . A_t^{\frac{-1}{1-\alpha}}\right)^{\varphi} \left(\left((1-\alpha)A_t^{\left(\frac{1+\varphi}{1-\alpha}\right)}\right)^{\frac{1-\alpha}{\varphi+(1-\alpha)\sigma+\alpha}}\right)^{\sigma}$$

Steady State

The deterministic steady state is defined as the equilibrium of the economy when the variance of the exogenous shocks is zero. In this state, variables are constant over time ($X_t = X_{t+1} = X$), and the technological shock is at its unconditional mean ($A = 1$). We also assume a zero-inflation steady state ($\pi = 0$).

- In steady state, the market clearing conditions imply:

$$A = 1 \mid Y = C \mid B = 0$$

Thus:

$$Y = (1-\alpha)^{\frac{1-\alpha}{\varphi+(1-\alpha)\sigma+\alpha}}$$

$$N = (1-\alpha)^{\frac{1}{\varphi+(1-\alpha)\sigma+\alpha}}$$

$$\frac{W}{P} = (1-\alpha)^{\frac{\varphi+(1-\alpha)\sigma}{\varphi+(1-\alpha)\sigma+\alpha}}$$

$$i = r = \frac{1-\beta}{\beta}$$

Log-linearization of the model

- Log-linearization is a technique used in macroeconomics to approximate non-linear models with linear versions that are easier to analyze.

$$\beta = \frac{1}{1+\rho}, Q_t = \frac{1}{1+i_t}, \pi_{t+1} = \frac{P_{t+1}}{P_t} - 1$$

Euler Equation

$$Q_t = E_t\left[\beta \frac{C_{t+1}^{-\sigma}}{C_t^{-\sigma}} . \frac{P_t}{P_{t+1}}\right]$$

$$C_t^{-\sigma} = E_t\left[\beta \frac{C_{t+1}^{-\sigma}}{Q_t} . \frac{P_t}{P_{t+1}}\right]$$

$$C_t^{-\sigma} = E_t[(1+i_t)(1+p)^{-1}(1+\pi_{t+1})^{-1}C_{t+1}^{-\sigma}]$$

Taking logs:

$$-\sigma \ln(C_t) = \ln(1+i_t) - \ln(1+p) - E_t \ln(1+\pi_{t+1}) - \sigma E_t \ln(C_{t+1})$$

Totally differentiating, we have:

$$-\sigma \frac{dC_t}{C} = \frac{di_t}{(1+i)} - \frac{d\pi_{t+1}}{(1+\pi)} - \sigma \frac{dC_{t+1}}{C}$$

$$\sigma E_t \tilde{c}_{t+1} - \sigma \tilde{c}_t = \tilde{\imath}_t - E_t \tilde{\pi}_{t+1}$$

$$\tilde{c}_t = E_t \tilde{c}_{t+1} - \frac{1}{\sigma}(\tilde{\imath}_t - E_t \tilde{\pi}_{t+1})$$

Labor Supply

$$\frac{W_t}{P_t} = C_t^{\sigma} N_t^{v}$$

$$\ln(W_t) - \ln(P_t) = \sigma \ln(C_t) + v \ln(N_t)$$

$$\frac{dW_t}{W} - \frac{dP_t}{P} = \sigma \frac{dC_t}{C} + v \frac{dN_t}{N}$$

$$\tilde{w}_t - \tilde{p}_t = \sigma \tilde{c}_t + v \tilde{n}_t$$

Demand of Labor

$$\frac{W_t}{P_t} = (1-\alpha) A_t N_t^{-\alpha}$$

$$\ln(W_t) - \ln(P_t) = \ln(1-\alpha) + \ln(A_t) - \alpha \ln(N_t)$$

$$\frac{dW_t}{W} - \frac{dP_t}{P} = \frac{dA_t}{A} + \alpha \frac{dN_t}{N}$$

$$\tilde{w}_t - \tilde{p}_t = \tilde{a}_t - \alpha \tilde{n}_t$$

Production Function

$$Y_t = A_t N_t^{1-\alpha}$$

$$\ln(Y_t) = \ln(A_t) + (1-\alpha)\ln(N_t)$$

$$\frac{dY_t}{Y} = \frac{dA_t}{A} + (1-\alpha)\frac{dN_t}{N}$$

$$\tilde{y}_t = \tilde{a}_t + (1-\alpha)\tilde{n}_t$$

Technological Shock

$$A_t = \rho_a A_{t-1} + \varepsilon_t^a$$

$$\frac{dA_t}{A} = \rho_a \frac{dA_{t-1}}{A} + d\varepsilon_t^a$$

$$\tilde{a}_t = \rho_a \tilde{a}_{t-1} + \varepsilon_t^a$$

Equilibrium in the Markets of Goods

$$Y_t = C_t$$

$$\ln(Y_t) = \ln(C_t)$$

$$\frac{dY_t}{Y} = \frac{dC_t}{C}$$

$$\tilde{y}_t = \tilde{c}_t$$

Monetary Policy and Price Level Determination

- All cases will make use of the Fisherian equation:

$$i_t = E_t[\pi_{t+1}] + r_t$$

Note: This implies that the nominal rate adjusts one for one with expected inflation, given a real interest rate that is determined exclusively by real factors.

Exogenous path the nominal interest rate

- Let us first consider the case in which the nominal interest rate follows an exogenous stationary path. Assume that i_t corresponds to a constant interest rate:

$$i_t = i = \rho$$

  In steady state with zero inflation and no secular growth.
- Rewriting the Fisherian equation:

$$E[\pi_{t+1}] = i_t - r_t$$

  - r_t is determined independently of the monetary policy.

$$p_{t+1} - p_t = i_t - r_t$$

$$p_{t+1} = p_t + i_t - r_t + \varepsilon_t$$

  - ε_t is a shock possibly unrelated to economic fundamentals and satisfying $E_t[\varepsilon_t] = 0$ for all t (sunspot shocks).

Note: An equilibrium in which such nonfundamental factors may cause fluctuations in one or more variables is referred to as indetermined equilibrium.

- The equilibrium path for the money supply is given by:

$$m_t - p_t = y_t - \eta i_t$$

  - Money supply is endogenous under the present policy regime.

Note: This relationship causes that the money supply will inherit the indeterminacy of p_t.

A Simple Inflation-Based Interest Rate Rule

- Suppose that the central bank adjusts the nominal interest rate according to the rule:

$$i_t = \rho + \phi_\pi \pi_t$$

where $\phi_\pi \geq 0$.

Combining with the Fisherian equation:

$$i_t = E_t[\pi_{t+1}] + r_t$$

$$\phi_\pi \pi_t = E_t[\pi_{t+1}] + \hat{r}_t$$

  - where $\hat{r}_t = r_t - \rho$.

Solving:

$$\pi_t = \frac{1}{\phi_\pi} E_t[\pi_{t+1}] + \frac{1}{\phi_\pi} \hat{r}_t$$

Take expectation at time t:

$$E[\pi_{t+1}] = \frac{1}{\phi_\pi} E_{t+1}[\pi_{t+2}] + \frac{1}{\phi_\pi} \hat{r}_{t+1}$$

Substituting back and after k iterations:

$$\pi_t = \sum_{k=0}^{\infty} \left(\frac{1}{\phi_\pi}\right)^{k+1} E_t[\hat{r}_{t+k}] + \lim_{k\to\infty} \left(\frac{1}{\phi_\pi}\right)^{k+1} E_t[\pi_{t+k+1}]$$

Case 1: $(\phi_{pi} > 1)$

- If $\phi_{pi} > 1$, the last term disappears:

$$\lim_{k\to\infty} \left(\frac{1}{\phi_\pi}\right)^{k+1} \to 0$$

Therefore, the unique bounded solution is:

$$\pi_t = \sum_{k=0}^{\infty} \left(\frac{1}{\phi_\pi}\right)^{k+1} E_t[\hat{r}_{t+k}]$$

Note: This condition for a determinate price level, requires that the central bank adjust nominal interest rates more than one for one in response to any change in inflation, a property known as the ***Taylor principle***.

Case 2: $(\phi_{pi} < 1)$

- The last term not disappears, then the steady stationary solution takes the form:

$$\pi_{t+1} = \phi_\pi \pi_t - \hat{r}_t + \varepsilon_t$$

where ε_t is, again, an arbitrary sequence of shocks.

- This allows self-fulfilling inflation paths (indetermination).

An Exogenous Path for the Monetary Supply

- Taking the log-linear form of the money demand:

$$m_t - p_t = y_t - \eta i_t$$

  - $\eta \geq 0$ denotes the interest semi-elasticity of money demad.

and replacing in the Fisher equation:

$$i_t = E_t[\pi_{t+1}] + r_t$$

$$\frac{1}{\eta}(y_t - m_t + p_t) = E_t[p_{t+1}] - p_t + r_t$$

$$p_t + \eta p_t = \eta E_t[p_{t+1}] + \eta r_t + \eta m_t$$

$$p_t = \left(\frac{\eta}{1+\eta}\right) E_t[p_{t+1}] + \left(\frac{\eta}{1+\eta}\right) m_t + \underbrace{\frac{1}{1+\eta}(\eta r_t - y_t)}_{u_t}$$

- Assuming $\eta > 0$ and taking expectation at time t:

$$E_t[p_{t+1}] = \left(\frac{\eta}{1+\eta}\right) E_{t+1}[p_{t+2}] + \left(\frac{\eta}{1+\eta}\right) m_{t+1} + u_{t+1}$$

Substituting back and after k iterations:

$$p_t = \frac{1}{1+\eta}\sum_{k=0}^{\infty}\left(\frac{\eta}{1+\eta}\right)^k E_t[m_{t+k}] + \underbrace{\sum_{k=0}^{\infty}\left(\frac{\eta}{1+\eta}\right)^k E_t[u_{t+k}]}_{u_t'}$$

- We can rewrite the before equation in terms of future growth of money supply:

$$p_t = m_t + \sum_{k=1}^{\infty}\left(\frac{\eta}{1+\eta}\right)^k E_t[m_{t+k}] + u_t'$$

***Note*:** An arbitrary exogenous path for the money supply always determines the price level uniquely.

- Given the price level, we can obtain the nominal interest rate:

$$i_t = \frac{1}{\eta}(y_t - m_t + p_t)$$

$$i_t = \frac{1}{\eta}\left(\sum_{k=1}^{\infty}\left(\frac{\eta}{1+\eta}\right)^k E_t[m_{t+k}] + u_t'\right) + \frac{1}{\eta}y_t$$

$$i_t = \frac{1}{\eta}\sum_{k=1}^{\infty}\left(\frac{\eta}{1+\eta}\right)^k E_t[m_{t+k}] + \frac{1}{\eta}(y_t + u_t')$$